



Inequality notation to express error

Task A: Range of values on a number line

1) Identify the **range** of unrounded values when:

a) 34 has been rounded to the nearest integer.

Smallest: 33.5

Largest: 34.49

b) 63 has been rounded to the nearest integer.

Smallest: 62.5

Largest: 63.49

c) 1 has been rounded to the nearest integer.

Smallest: 0.5

Largest: 1.49

2) Identify the **range** of unrounded values when:

a) 6 has been rounded to 1 s.f.

Smallest: 5.5

Largest: 6.49

b) 56 has been rounded to the 2 s.f.

Smallest: 55.5

Largest: 56.49

c) 1000 has been rounded to 4 s.f.

Smallest: 999.5

Largest: 1000.49

3) Work out the question given the following:

a) 4 has been rounded to 1 s.f.

Smallest: 3.5

Largest: 4.49

b) 98 has been rounded to the 2 s.f.

Smallest: 97.5

Largest: 98.49

c) 100 has been rounded to the nearest integer.

Smallest: 99.5

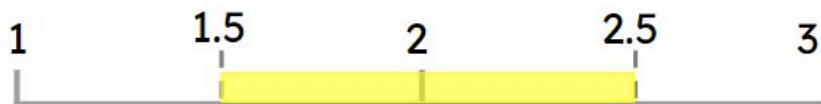
Largest: 100.49

Task B: Error intervals

1) Given the number lines, identify the **error interval**.

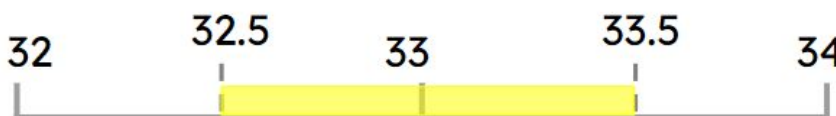
a) 2 rounded to 1 s.f.

$$1.5 \leq x < 2.5$$



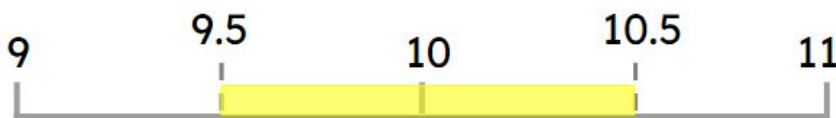
b) 33 rounded to 2 s.f.

$$32.5 \leq x < 33.5$$



c) 10 rounded to 2 s.f.

$$9.5 \leq x < 10.5$$



2) Identify the error interval of the following:

a) 482 is rounded to the nearest whole number $481.5 \leq x < 482.5$

b) 65 is rounded to the 2 s.f.

$$64.5 \leq x < 65.5$$

c) 7.2 is rounded to 1 d.p.

$$7.15 \leq x < 7.25$$

d) 4.54 is rounded to 2 d.p.

$$4.535 \leq x < 4.545$$

3) Pair up the question with the correct **error interval**

